REGULARITY OF LOCALISATIONS IS PRESERVED BY ETALE RING MAPS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The localisations are Noetherian after an etale extension. Lemma \ref{lemma-Noetherian-etale-extension}, cited at proof lines 10865-10866. Justifies assuming both local rings are Noetherian.
- Faithful flatness of an etale map at a prime, and the behaviour of minimal generating sets of maximal ideals. Algebra, Lemma \ref{algebra-lemma-etale-at-prime}, cited at proof lines 10871-10873. Shows the images of a minimal generating set form a minimal generating set.
- Etale maps preserve the dimension of the localisations at corresponding primes. Lemma \ref{lemma-dimension-etale-extension}, cited at proof lines 10876-10877. Supplies the equality of dimensions used in the definition of regularity.

Source reference. Original lemma 0AH0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13213–13213. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0AGZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13212–13212. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00U4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1098–1098. Navigation pointer to a statement the

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Source reference. Original lemma 07QP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9839–9839. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *If $A \to B$ is an étale ring map and $\mathfrak{q}$ is a prime of B lying over $\mathfrak{p} \subset A$, then $A_\mathfrak{p}$ is regular if and only if $B_\mathfrak{q}$ is regular.*

Proof. By Lemma 0AGZ we may assume both $A_\mathfrak{p}$ and $B_\mathfrak{q}$ are Noetherian in order to prove the equivalence. Let $x_1, \ldots, x_t \in \mathfrak{p}A_\mathfrak{p}$ be a minimal set of generators. As $A_\mathfrak{p} \to B_\mathfrak{q}$ is faithfully flat we see that the images $y_1, \ldots, y_t$ in $B_\mathfrak{q}$ form a minimal system of generators for $\mathfrak{p}B_\mathfrak{q} = \mathfrak{q}B_\mathfrak{q}$ (Algebra, Lemma 00U4). Regularity of $A_\mathfrak{p}$ by definition means $t = \dim(A_\mathfrak{p})$ and similarly for $B_\mathfrak{q}$. Hence the lemma follows from the equality $\dim(A_\mathfrak{p}) = \dim(B_\mathfrak{q})$ of Lemma 07QP. □